

2.3 Emerald Rapids: 5th-Generation Intel® Xeon® Scalable Processors

Ashley O. Munch¹, Nevine Nassif¹, Carleton L. Molnar¹, Jason Crop², Rich Gammack¹, Chinmay P. Joshi³, Goran Zelic¹, Kambiz Munshi¹, Min Huang⁴, Charles R. Morganti², Sireesha Kandula¹, Arijit Biswas¹

¹Intel, Hudson, MA

²Intel, Fort Collins, CO

³Intel, Hillsboro, OR

⁴Intel, Santa Clara, CA

The 5th-Generation Intel Xeon Scalable Processors (codenamed Emerald Rapids) support up to 64 cores, greater than 300MB shared L3 cache, 8 DDR5 channels at 5600MT/s with 1DPC, 32GT/s PCIe/CXL lanes, 20GT/s UPI lanes composed of 2 die (Fig. 2.3.1) in a multichip package. This generation delivers an 18% performance improvement for general integer compute workloads and a 24% improvement for floating-point workloads at iso power vs. the 4th-Gen Xeon processors [1]. This is achieved through an improved core, enhanced process technology, increased core count, significantly larger cache, higher DDR memory speeds, die reductions, and power efficiency improvements at idle conditions. Manufacturing is on the Intel 7 process technology optimized for server usage with improved transistor speed and a focus on leakage and dynamic capacitance reduction enabling +3% frequency/W enhancements over the prior generation process revision. The key power-efficiency improvements at idle were achieved through improvements to the fully integrated voltage regulators (FIVR) [2, 3] to decrease the regulator losses at lower utilization, enhanced active idle detection and power savings, and package C6 power reduction.

The maximum configuration (XCC) is composed of 2 die with one being a physical mirror of the other (referred to as XNC and XSC in Fig. 2.3.7). Each die is built as a 7×7 array of modular integration components that are connected with an on-die scalable coherent fabric (SCF) to provide low-latency, high-bandwidth communication throughout the array. Each modular integration component contains a SCF agent which provides access on and off of the fabric, this includes the core tiles (core plus cache, Fig. 2.3.3), memory controllers (MC), PCIe/CXL agents, UPI agents, and on-die cryptographic, compression and data streaming accelerators. The MC occupies a slot in the first and last columns of the array adjacent to the associated DDR PHY with the remaining slots populated with core tiles. To enable full bandwidth to each I/O agent, two dedicated interleaving rows of SCF are used along the northcap of the die allowing parallel paths to/from the array of cores and memory (Fig. 2.3.2). On-die voltage regulators (FIVRs) are arranged in horizontal strips under each row in nearest proximity to the circuitry they are powering.

The XNC and XSC die are connected by a total of 3 Embedded Multi-Die Interconnect Bridges (EMIBs), which are used to extend the on-die SCF cross die. Each EMIB is embedded into the package substrate consuming 2 package routing layers and blocks any direct package to silicon connectivity. As such, all power delivery to circuitry in the shadow of the EMIB must be done by package surface layer, passthrough the EMIB, or cantilever distribution on the CPU die. Due to reliability constraints in packaging and assembly, the length and aspect ratio of each EMIB is limited requiring multiple bridges on each vertical and horizontal die-crossing edge. These gaps between EMIBs are used for FIVRs providing power to the adjacent circuitry. The result is a quasi-monolithic SoC, consisting of a total of 66-printed cores with one core on each die used for redundancy.

The number of cores and cache were extended beyond those of the prior generation, while reducing the number of die and total silicon area in the package by merging the north and south quadrants into hemispheres, which reduced area, power and latency needed to support die-to-die crossing and allowed optimization of redundant components which was not possible with a 4-identical quadrant design [1]. While the individual die are larger relative to a 4-die configuration, the decreased manufacturing defect density and the elimination of multiple die-to-die crossings results in lower total packaging, assembly, and test (PAT) cost.

The Last Level Cache (LLC) is a unified cache architecture shared across all cores with a total size of over 300MB. It is physically constructed as 5MB per core tile with 20 ways and 4K sets (Fig. 2.3.3). The size represents an increase of over 2.5× compared to the previous generation at a 30% higher density without increasing the pipeline latency at iso-frequency. This was achieved through a combination of larger memory array instances with greater area efficiency, LLC level floorplan optimizations, and improved logic area utilization. The LLC is accompanied by a 3.75MB Local Snooper Filter (LSF) with 15 ways and 4K sets and a 1MB Remote Snooper Filter (RSF) with 4 ways and 4K sets. All the arrays are built with 6T SRAMs cells and include read- and write-assist circuits for improved low-voltage operation. Leakage power is minimized through standard array sleep techniques with less than 2% of the data arrays woken up for an

access. To reduce Soft Error Rate (SER) impacts, the LLC data arrays are protected by DECTED ECC, while all other arrays are covered with SECDED ECC. Centralized repair with row, column, and bit coverage is utilized on the LLC data arrays to reduce the area overhead and enhance the repair capability; traditional row and column repairs are used in the remaining arrays. The combination of the read/write-assist circuits, ECC coverage, and advanced repair schemes enable a very low operating voltage for the LLC.

The 5th-Gen Xeon processors support 8 channels of DDR5 memory at 5600 MT/s with 1 DIMM per channel (1DPC). Multiple circuit optimizations including MIM capacitance, package routing, and on-die power delivery improvements along with functional optimizations, such as increased Tx/Rx FIFOs depths and finer gain controls of the strobe receiver, enabled the increase to 5600 MT/s over the prior generation. On-die local voltage regulators provide quiet supplies (Vddrldo) to the Tx and Rx clocking circuits (Fig. 2.3.4) to minimize power-supply noise induced jitter primarily for READ and partially for WRITE operations, while allowing for advanced features to mitigate supply droop and enable low Tx jitter and Rx margin (Fig. 2.3.5). The increase in memory speed, together with increased core count and larger LLC increased overall memory bandwidth by 1.14× per socket and 1.2× inter-socket compared with the prior generation.

To ensure robust and reliable operation across all conditions, the design was optimized to specifically account for the effects exacerbated in large multichip systems. This included package power delivery improvements to minimize variation along with design optimizations to cover static voltage gradients, work-load dependent dynamic droops, and on-die thermal gradients (Fig. 2.3.6). Global clock uncertainty charges were minimized by crediting uniform design topology, and to ensure timing robustness of the global clocks careful consideration was made for metal layer-to-layer variation, accurately extracted capacitance in context of top-side metal, and accounting for across-wafer systematic device variation. Nonuniformly designed local clock timing margins were used to account for silicon to simulation miscorrelation that was observed to vary between gate type, size, and loading. Modular Die Fabric Interconnect (MDFI) clock duty cycle was controlled by intermittent correction at the phase-locked loop (PLL) and guard banded for residual aging. With power and performance optimization a key objective, the clock distribution was designed to balance tight skew requirements through the use of robust power efficient design and circuit techniques. This included optimizing the infrastructure clock distribution across the entire die to minimize the number of global drop-off points and changing from a wide to shorter recombination spine topology for high-speed clocks, while still achieving tight skew that reaped up to 20% savings in the global clock distribution power. Even with the physically larger die, low jitter requirements for the LCPLLs in UPI and PCIe were met with differential clock distribution using top metal and additional specialized clock buffers in the distribution.

Acknowledgement:

The authors thank all the teams for their uncompromising dedication in bringing this product to market.

References:

- [1] N. Nassif et al., "Sapphire Rapids: The Next-Generation Intel Xeon Scalable Processor" *ISSCC*, pp. 44-45, 2022.
- [2] E. A. Burton et al., "FIVR – Fully Integrated Voltage Regulators on 4th Generation Intel Core SoCs," *IEEE APEC*, pp. 432-439, 2014.
- [3] K. Bharath et al., "Integrated Voltage Regulator Efficiency Improvement using Coaxial Magnetic Composite Core Inductors," *IEEE Elec. Components and Tech. Conf.*, pp. 1286-1292, 2021.

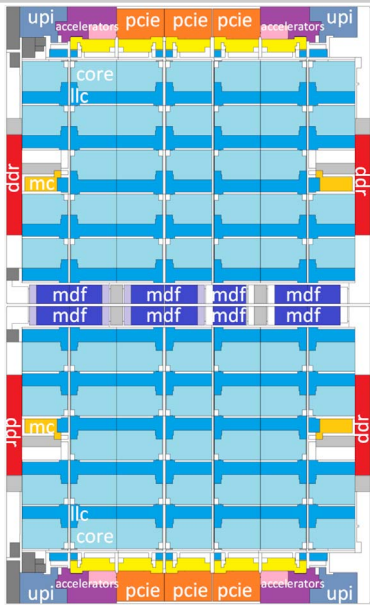


Figure 2.3.1: Processor floorplan diagram for 2-die XCC configuration.

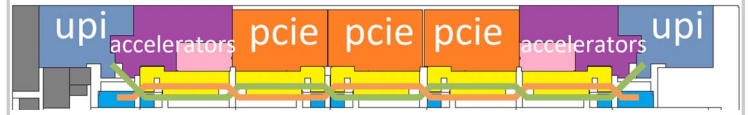


Figure 2.3.2: I/O northcap with 2 interleaving rows of SCF interconnect.

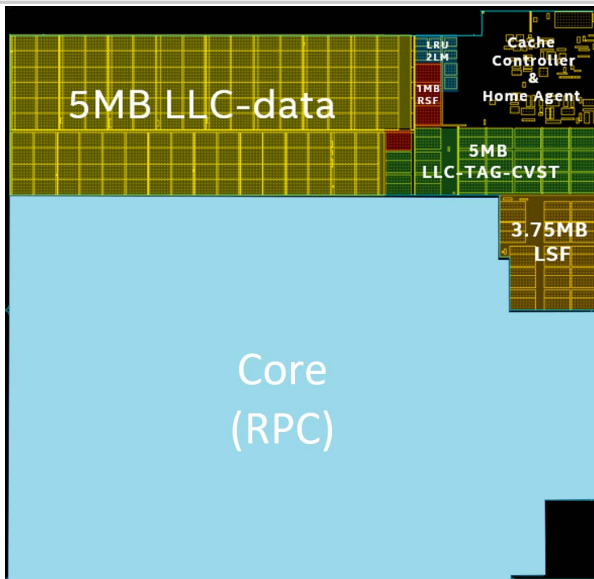


Figure 2.3.3: Core tile floorplan with Core, 5MB LLC cache, 3.75MB Local Snoop Filter (LSF) and a 1MB Remote Snoop Filter (RSF).

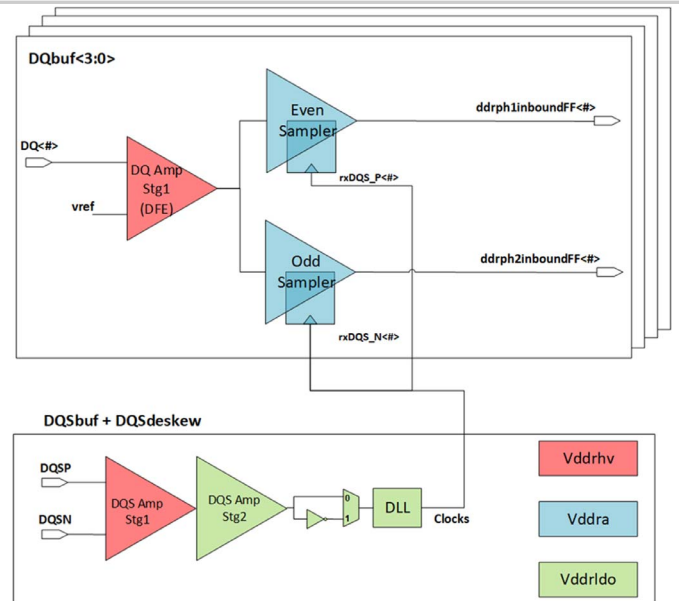


Figure 2.3.4: Voltage domains for DDR receiver data path and clock circuit.

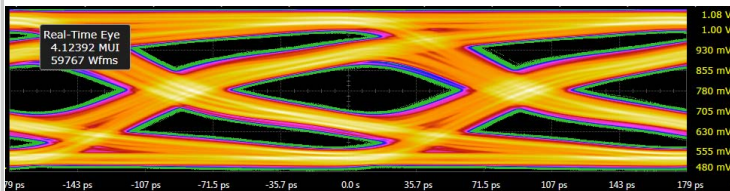


Figure 2.3.5: Silicon measured DDR lane eye diagram at 5600MT/s.

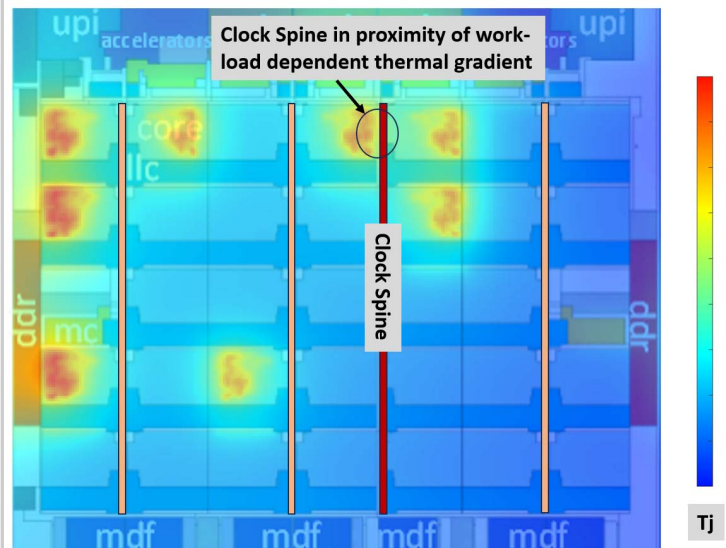


Figure 2.3.6: Illustration of one of the many work-load dependent on-die thermal gradient cases compensated for in design.

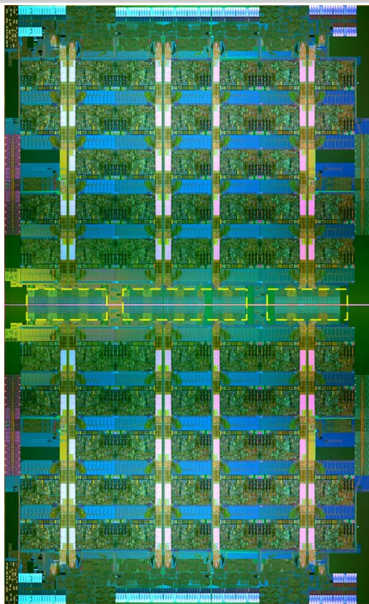


Figure 2.3.7: XNC and XSC die photograph (with EMIB locations highlighted).